Sebastian "Sebo" Diaz

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OVERVIEW

- HST MIT PhD candidate in RLE & CSAIL.
- Research interests in machine learning, (convex) optimization, representation learning, generalization, data augmentation, and reinforcement learning.

EDUCATION

Massachusetts Institute of Technology — 2023 – May 2028 (expected)
Ph.D. HST (Computer Science) — GPA: 5.00 / 5.00

University of Arizona — 2019 – 2023
B.S. in Biomedical Engineering; Minor in Mathematics — GPA: 3.95 / 4.00

INDUSTRY EXPERIENCE

Machine Learning Research, Stealth Startup — May 2026 – Sept. 2026
Kendall Square, Cambridge, MA

- Post-training V/LLM research in non-verifiable RL settings.

FELLOWSHIPS & AWARDS

Graduate Research Fellowship Program, National Science Foundation — 2024
MathWorks Graduate Fellow, MathWorks — 2024
School of Engineering – Distinguished Graduate Student Fellow, MIT — 2023
Program for Exemplary Mentoring Fellow, MIT — 2023
Goldwater STEM Scholarship, Goldwater Foundation — 2022
Sophomore of the Year, University of Arizona — 2021
Native Health Care Scholar, Udall Foundation — 2021

PUBLICATIONS

Lead Author

1. **Sebastian Diaz**, Polina Golland, Elfar Adalsteinsson, Neel Dey. “Why Invariance is Not Enough for Biomedical Domain Generalization and How to Fix It”. *Preprint: arXiv:2604.02564*, 2026.
2. **Sebastian Diaz**, Benjamin Billot, Neel Dey, Mingxuan Zhang, Esra Abaci-Turk, Ellen Grant, Polina Golland, Elfar Adalsteinsson. “Robust Fetal Pose Estimation across Gestational Ages via Cross-Population Augmentation”. *Medical Image Computing and Computer Assisted Intervention (MICCAI)*, 2025.
3. **Sebastian Diaz**, Yamin Arefeen, Borjan Gagoski, Ellen Grant, Elfar Adalsteinsson. “Design of Novel RF Pulse for Fetal MRI Refocusing Trains Using Rank Factorization (SLfRank) to Reduce SAR and Improve Image Acquisition Efficiency”. *International Society for Magnetic Resonance in Medicine (ISMRM)*, 2022.

Contributing Author

1. Yingcheng Liu, Peiqi Wang, **Sebastian Diaz**, Benjamin Billot, Esra Abaci-Turk, Ellen Grant, Polina Golland. “Fetuses Made Simple: Modeling and Tracking of Fetal Shape and Pose”. *Medical Image Computing and Computer Assisted Intervention (MICCAI)*, 2025.

2. Molin Zhang, Nicolas Arango, **Sebastian Diaz**, Jacob White, Elfar Adalsteinsson. “Stochastic-Offset-Strategy Enhanced RF Pulse Optimization with Auto-Differentiation”. *International Society for Magnetic Resonance in Medicine (ISMRM)*, 2024.
3. Ricky Cordova, Kelli Kiekens, Susan Burrell, William Drake, Zaynah Kmeid, Photini Rice, Andrew Rocha, **Sebastian Diaz**, Shigehiro Yamada, Michael Yozwiak, Omar L. Nelson, Gustavo C. Rodriguez, John Heusinkveld, Ie-Ming Shih, David S. Alberts, Jennifer K. Barton. “Sub-Millimeter Endoscope Demonstrates Feasibility of In Vivo Reflectance Imaging, Fluorescence Imaging, and Cell Collection in the Fallopian Tubes”. *Journal of Biomedical Optics*, 2021.

INVITED TALKS

RepreSeg: Stable Representations for Domain Generalized Segmentation
MIT IMES Seminar, Cambridge, MA — Dec. 12, 2025

Learning Alongside Invariant Representations: Robust Generalization in Medical Imaging
MIT IMES Department Retreat, Boston, MA — Sept. 29, 2025

RELEVANT COURSEWORK

MIT	Deep Learning, Numerical Methods, Computer Vision, fMRI Neuroscience
Harvard Medical School	Pathology, Immunology, Genetics, Cardiovascular & Pulmonary Pathophysiology

SKILLS

Programming	Python, PyTorch, C++
Technical	Numerical optimization, probability

SERVICE

Reviewer, MICCAI: PIPPI	2025–26
Reviewer, CVPR	2026
Reviewer, NeurIPS	2026